# TONER-1-V

**363,568 parameters**
260,032 story + 103,536 doodle, int8, one scale per row
Inference runs inside the PostScript interpreter

State of the art among multimodal models that execute inside a print job.

### SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little boy named Tim. Tim loved to play outside with his friends. They would make more honey and run around the house. One day, Tim and his friends played together every day. Tim was happy because he didn't have a safe explorer of the tape.

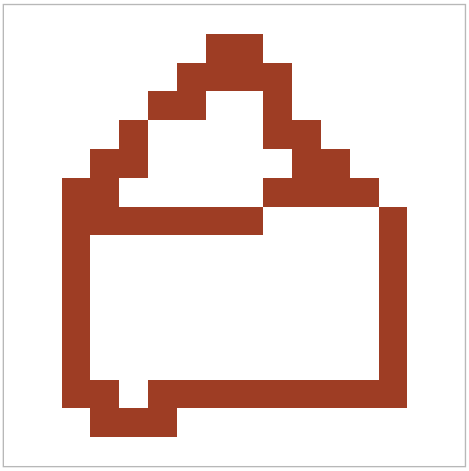
Tim and his friends had an idea. They decided to take him a bike out the storm. They found the bikes and found a big box. When it was waiting, they did not like to make it anymore. Tim was scared, but…

### ILLUSTRATION

Drawn after the story by TONER-1-V, in the same print job, from 64 patch tokens.

### PATCH TOKENS

```
. . . 1 2 . . .
. . 1 e d . . .
. 1 a . 4 b . .
. f 3 3 6 c 9 .
. a . . . . 5 .
. a . . . . 5 .
. b 1 3 3 3 7 .
. 4 c . . . . .
```



*TONER-1-V's drawing of a house: 16 × 16 pixels, generated inside this print job, upscaled by proprietary toner super-resolution.*

### CLASS ROUTER

First drawable noun in the story: "house", word 29 of 94. Class token 29 (house).

### SAMPLING

BOS, the class token, then 64 patch tokens at temperature 0.9, each drawn from the 16 patch values only. 65 forward passes.

| STORY · stories260K | | | | DOODLE · TONER-1-V | | | |
|---|---|---|---|---|---|---|---|
| TOKENS | TOK / SEC | SEED | TEMP | TOKENS | TOK / SEC | SEED | TEMP |
| 200 | 63.7 | 7 | 0.8 | 64 | 206 | 7 | 0.9 |

| | |
|---|---|
| **INTERPRETER** | GPL Ghostscript · revision 10071 · PostScript version 3010 |
| **STORY MODEL** | stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab |
| **DOODLE MODEL** | TONER-1-V (Llama-2 architecture) · dim 48 · 4 layers · 6 query / 3 KV heads · 36-token vocab |
| **STOPPED BY** | Token budget: STEPS = 200. The story was cut off mid-thought. · doodle: 64 patch tokens, fixed length |

**Host compute: 0 FLOPs. Inference cost includes toner, now in color.**

Every word and every pixel above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seeds on another interpreter tell a different story and draw a different picture. Drawings learned from the Quick, Draw! dataset (Google, CC BY 4.0).